Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/20/2025
PI Name:	Baker, John Ph.D.		
Project Title:	Defining the Relationship Between Simulated Weightlessness and Space Radiation on Cardiovascular Disease, and the Effectiveness of a Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53226-3548	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	02/04/2019	**End Date:**	02/03/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/03/2024 per NSSC information (Ed., 4/18/23) NOTE: End date changed to 02/03/2023 per NSSC information (Ed., 11/3/22)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Kronenberg, Amy Sc.D. (Lawrence Berkeley National Laboratory) Willey, Jeffrey Ph.D. (Wake Forest University)		
Grant/Contract No.:	80NSSC19K0498		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The interaction between simulated weightlessness and space radiation on cardiac disease needs to be understood prior to undertaking deep space exploratory missions. To date, combining the simulated spaceflight hazards of reduced weight bearing via hind limb unloading in rats of any strain with simultaneous GCR or gamma exposure in conscious animals has not been possible to conduct at the NASA Space Radiation Laboratory. To meet this challenge, we have worked with Jeffrey Willey, PhD, our collaborator at Wake Forest Medical School, and have engineered a solution that provides a realistic scenario for the investigation of exploratory spaceflight. Thus, our current ground-based studies directly address this important aspect of the NASA Human Research Program Degenerative Tissue Gaps as set forth in the Human Research Roadmap. The research currently underway is providing important information to help close these gaps. We determined the combined effects of simulated weightlessness and radiation exposure during deep space exploration on risk for and occurrence of cardiac disease in a ground-based rat model. These studies simulate a 3-3.5-year deep space human mission in male WAG/RijCmcr rats 6 months old at the start of the study, determining the impact of simulated weightlessness combined with components of GCRs using a simplified 5-ion mixed field beam with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for cardiac disease. Parallel studies were conducted with male rats from a second strain (Fischer 344) at an age representative of the astronaut corps. Specific Aim A: Simulate a 3-3.5 year deep space human mission in male WAG rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease. Specific Aim B: Determine the response to photon radiation in male Fischer 344 rats with and without simulated weightlessness to estimate relative biological effectiveness (RBE) for early clinically relevant endpoints of risk for and occurrence of cardiac disease. Rats will be irradiated with 3 doses of X-rays: 3.0 Gy, 8.0 Gy, and 10.0 Gy alongside experimental hindlimb unloaded animals. Specific Aim C: Simulate a 3-3.5 year deep space human mission in male Fischer 344 rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth as it relates to patients who receive therapeutic radiation for the treatment of cancer and to individuals who receive accidental exposure to radiation from a nuclear accident.
Task Progress:	In WAG rats, our findings provide evidence for the occurrence of cardiac fibrosis resulting from exposure of rats to a 5-ion mixed field beam designed to broadly simulate the composition of the galactic cosmic rays (GCR) behind shielding, with no additive effect of the simulated spaceflight hazard of reduced weight bearing via hindlimb unloading (HLU) on the impact of simplified Galactic Cosmic Ray simulation (SimGCRSim). These findings were presented at the NASA Human Research Program Investigators' Workshop (IWS) in 20924. Fischer 344 rats did not show an increase in blood urea nitrogen, cholesterol, or cardiac perivascular collagen content after exposure to SimGCR and X-radiation with no impact of simulated weightlessness on the phenotypes measured. Characterizing the biological response of animal models to radiation and understanding the resulting biological damage is essential in establishing models of risk and developing effective countermeasure strategies to enable safe, long-duration space travel. Ground-based animal studies have been used to assess the increased risk for developing degenerative cardiovascular disease. The in vivo rat model enables the effective, reproducible, and economic study of cardiovascular function, metabolism, and ultrastructure over a wide range of experimental conditions, providing detailed physiological characterization that can be studied further using molecular genetic tools. Inbred rat strains may be useful in the development of better physiological and biochemical models to study radiosensitivity, because these strains should yield less inter-animal variation, and environmental variables can be controlled. For the study of radiosensitivity to ionizing radiation, the whole body can be readily subjected to radiation. Thus, rats from two genetically distinct inbred strains were exposed to the same doses of radiation and cardiovascular function determined non-invasively in longitudinal follow up studies. Our findings suggest that WAG and F344 rats may be suitable as models to study susceptibility and resistance to radiation, respectively. It is likely that the strain differences we observed have a genetic component; therefore, these genetic components can be defined and used to help identify environmental components, including simulated weightlessness and the stress of social isolation that are also likely to play a role in radiosensitivity. In conclusion, molecular genetic strategies can be used to further our understanding of radiosensitivity and lead to better strategies to confer protection to astronauts during extended exploratory missions.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
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